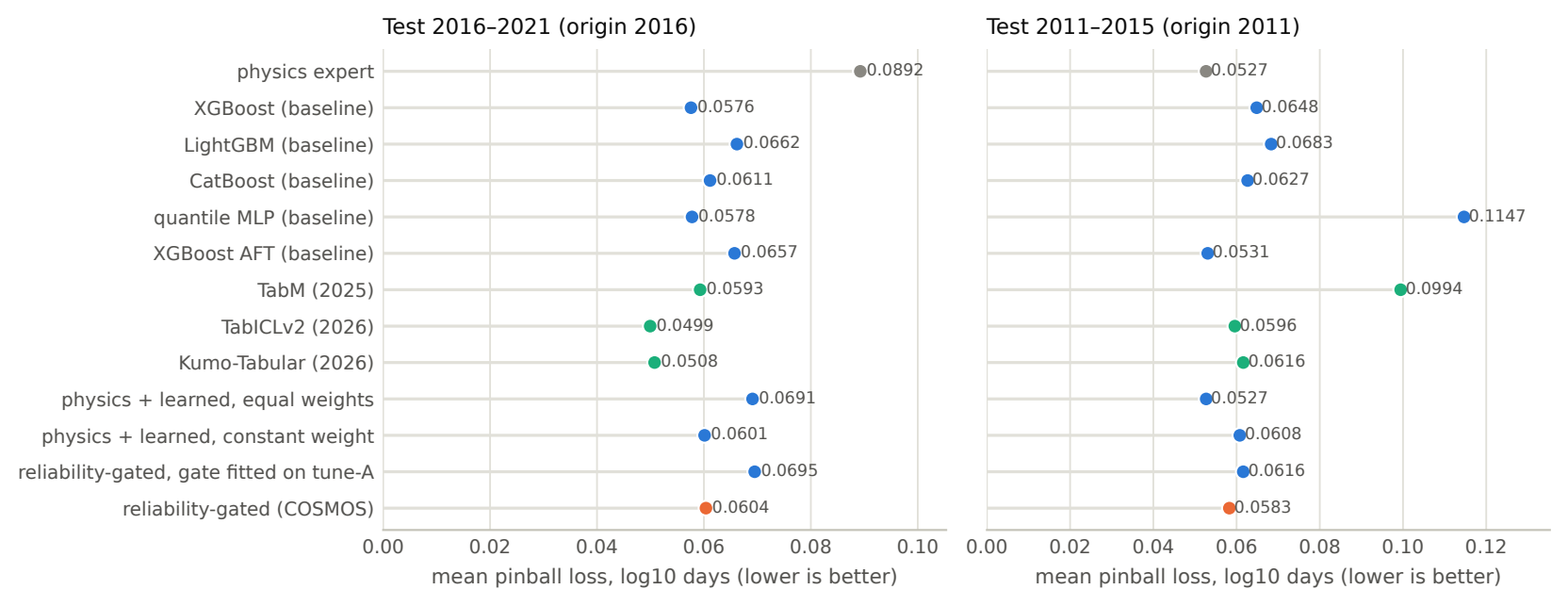
Test 2016–2021 (origin 2016)
physics expert
XGBoost (baseline)
LightGBM (baseline)
CatBoost (baseline)
quantile MLP (baseline)
XGBoost AFT (baseline)
TabM (2025)
TabICLv2 (2026)
Kumo-Tabular (2026)
physics + learned, equal weights
physics + learned, constant weight
reliability-gated, gate fitted on tune-A
reliability-gated (COSMOS)
0.0892
0.0576
0.0662
0.0611
0.0578
0.0657
0.0593
0.0499
0.0508
0.0691
0.0601
0.0695
0.0604
0.00
0.02
0.04
0.06
0.08
0.10
mean pinball loss, log10 days (lower is better)
Test 2011–2015 (origin 2011)
0.0527
0.0648
0.0683
0.0627
0.1147
0.0531
0.0994
0.0596
0.0616
0.0527
0.0608
0.0616
0.0583
0.00
0.02
0.04
0.06
0.08
0.10
0.12
mean pinball loss, log10 days (lower is better)